Physics 468/668: Statistical Physics **Spring 2019**

Lecture 7: The Canonical Ensemble

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7.1 Background

We established the basis of ensemble theory and made a detailed study of the micro-canonical ensembles in chapter 2. In that ensemble, the macrostate of the systems was defined through a fixed (N, V, E), where we allow E has a fluctuation of Δ. The basic problem is to determine $\Omega(N, V, E, \Delta)$. For some systems, the complete thermodynamics can be derived.

The concept of a fixed energy (even with a range) is not very practical. First, the total energy E of a system is hardly to be measured. Second, it is hard to keep its value under strict physical control. In this sense, it is more practical to speak of a fixed temperature. Such an ensembel is referred to as a canonical ensemble.

In canonical ensemble, the energy E of a system variable,. It can take any value between 0 and infinity. The consequent question is, *what is the probability P_r that, at any time, a system is found to be one of the state characterized by the energy value E_r?*

There are two ways to determine the relation.

1. one consists of regarding the system as in equilibrium with a heat reservior at a common T
2. one consists of regarding the system as in a member of a canonical ensemble (N, V, T) in contact with many others.

We expect that in the thermodynamic limit the final results would be the same in both cases. Once P_r is determined, the rest follows without difficulty.

7.2 Equilibrium between a system and a heat reservoir

In a heat reservoir, the system and the reservoir would have a common T. Their energies, however, would be variable, in principle could have, at any time, values between 0 and $E^{(0)}$, where $E^{(0)}$ denotes the energy of the composite system $A + A'$. If the system A happens to be in a state characterized by the energy value E_r, the reservoir would have an energy E'_r, such that

$$E_r + E'_r = E^{(0)} = \text{const} \tag{7.1}$$

We assume that the reservoir is much larger than the given system, thus we have

$$\frac{E_r}{E^{(0)}} = \left(1 - \frac{E'_r}{E^{(0)}}\right) \ll 1 \tag{7.2}$$

When the state of A (E_r) is specified, the reservoir A′ can still be in any one of a large number of states compatible with the energy value E'_r. Let the number of these states be denoted by $\Omega'(E'_r)$. The prime on the symbols emphasizes the fact that its functional form will depend on the reservoir. Since the various possible states are equally likely to occur, the relevant probability would be directly proportional to this number.

$$P_r \propto \Omega'(E'_r) = \Omega'(E^{(0)} - E_r) \tag{7.3}$$

Expanding the logarithm of $\Omega'(E'_r)$,

$$\ln \Omega'(E'_r) = \ln \Omega'(E^{(0)}) + \left(\frac{\partial \ln \Omega'}{\partial E'}\right)_{E'=E^{(0)}} (E'_r - E^{(0)}) + ... \approx \text{const} - \beta' E_r \tag{7.4}$$

where we defined β in the previous lecture,

$$\beta = \left(\frac{\partial \ln \Omega}{\partial E}\right)_{N,V} \tag{7.5}$$

Note that, in equilibrium, $\beta = \beta' = 1/kT$. From the above equations, we obtain the desired results,

$$P_r \propto \exp(-\beta E_r) \tag{7.6}$$

After normalization, we get

$$P_r = \frac{\exp(-\beta E_r)}{\sum_r \exp(-\beta E_r)} \tag{7.7}$$

7.3 A system in the canonical ensemble

We consider an ensemble of N identical systems (0,1,2,...N) sharing a total energy E; let E_r denote the energy eigenvalues of the systems. If n_r denotes the number of systems which, at any time t, have the energy value E_r, then the set of numbers n_r must satisfy the conditions,

1. n_1 systems have energy of E_1
2. n_2 systems have energy of E_2
3. n_3 systems have energy of E_3
4. .....

$$\sum_r n_r = N \tag{7.8}$$

$$\sum_r n_r E_r = E = NU \tag{7.9}$$

where U denotes the average energy per system in the ensemble. Any set n_r that satisfies the above conditions is a possible state. Such state can be realized in a number of ways by permutation, therefore, the total number is

$$W\{n_r\} = \frac{N!}{n_0! n_1! \cdots} \tag{7.10}$$

All possible states are equally likely to occur, the frequency with a given $\{n_r\}$ will be directly proportional to the number $W\{n_r\}$. With a very large number of particles, we will find the distribution of $W\{n_r\}$ is very localized. That is, the most probable distribution of $\{n_r\}$ which lead to the maximum W is the only possible state.

7.4 The most probable values

In order to maximize W, we choose to work with $\ln W$,

$$\ln W\{n_r\} = \ln N! - \sum_r \ln n_r! \tag{7.11}$$

To the limit $N \to \infty$, the values of n_r would also tend to infinity. Applying the Stirling approximation,

$$\ln W\{n_r\} = N\ln N - \sum_r n_r \ln n_r \tag{7.12}$$

If we shift from the set $\{n_r\}$ to a slightly different set $\{n_r + \delta n_r\}$, the expression would change to

$$\delta(\ln W\{n_r\}) = -\sum_r \delta n_r(\ln n_r + 1) \tag{7.13}$$

If the set $\{n_r\}$ is maximal, the variation $\delta(\ln W\{n_r\})$ should vanish. At the same time, it also satisfies,

$$\sum_r \delta n_r = 0 \tag{7.14}$$

$$\sum_r \delta n_r E_r = 0 \tag{7.15}$$

Therefore, the desired set $\{n_r\}^*$ is determined by the method of Lagrange multipliers, by which the condition becomes,

$$\sum_r (-(\ln n_r^* + 1) - \alpha - \beta E_r)\delta n_r = 0 \tag{7.16}$$

Where α and β are coefficients due to Lagrange multipliers. The only way to satisfy the condition is to let all coefficients vanish identically,

$$\ln n_r^* = -(\alpha + 1) - \beta E_r \tag{7.17}$$

which gives

$$n_r^* = C\exp(-\beta E_r) \tag{7.18}$$

To determine C and β, we subject the equation to the original condition,

$$\frac{n_r^*}{N} = \frac{\exp(-\beta E_r)}{\sum_r \exp(-\beta E_r)} \tag{7.19}$$

The parameter β should be a solution of the equation,

$$U = \frac{\sum_r \exp E_r(-\beta E_r)}{\sum_r \exp(-\beta E_r)} \tag{7.20}$$

7.5 The mean values

To evaluate this expression, we replace the weight factors

$$\tilde{W}\{n_r\} = \frac{N! w_0^{n_0} w_1^{n_1} w_2^{n_2} \cdots}{n_0! n_1! n_2! \cdots} \tag{7.21}$$

$$\Gamma(N,U) = \sum{}' \tilde{W}\{n_r\} \tag{7.22}$$

TODO

7.6 Statistic quantities in the canonical ensemble

With the canonical distribution, we now look for a general recipe to extract information about the various macroscopic properties of the given system. Remember the definition of Helmholtz free energy $A = U - TS$

$$dA = dU - TdS - SdT = -SdT - PdV + \mu dN \tag{7.23}$$

$$S = -\left(\frac{\partial A}{\partial T}\right)_{N,V} \tag{7.24}$$

$$P = -\left(\frac{\partial A}{\partial V}\right)_{N,T} \tag{7.25}$$

$$\mu = -\left(\frac{\partial A}{\partial N}\right)_{V,T} \tag{7.26}$$

and

$$U = A + TS = A - T\left(\frac{\partial A}{\partial T}\right)_{N,V} = -T^2\left[\frac{\partial}{\partial T}\left(\frac{A}{T}\right)\right]_{N,V} = \left[\frac{\partial (A/T)}{\partial (1/T)}\right]_{N,V} \tag{7.27}$$

Comparing this equation with

$$U = \frac{\sum_r E_r \exp(-\beta E_r)}{\sum_r \exp(-\beta E_r)} = -\frac{\partial}{\partial \beta} \ln\left[\sum_r \exp(-\beta E_r)\right] \tag{7.28}$$

we find there is a close relation that

$$-\frac{A}{kT} = \ln\left[\sum_r \exp(-\beta E_r)\right] \tag{7.29}$$

The equation gives that

$$A(N,V,T) = -kT \ln Q_N(V,T) \tag{7.30}$$

where $Q_N(V,T)$ is the partition function of the system.

$$Q_N(V,T) = \sum_r \exp(-E_r/kT) \tag{7.31}$$

Once A is known, one could derive the rest quantities such as S, P, μ, and C_V as well,

$$C_V = \left(\frac{\partial U}{\partial T}\right)_{N,V} = -T\left(\frac{\partial^2 A}{\partial T^2}\right)_{N,V} \tag{7.32}$$

$$G = A + PV = A - V\left(\frac{\partial A}{\partial V}\right)_{V,T} = N\left(\frac{\partial A}{\partial N}\right)_{V,T} = N\mu \tag{7.33}$$

The entropy is the system is determined by $P_r = Q^{-1}\exp(-\beta E_r)$

$$< \ln P_r >= -\ln Q - \beta < E_r >= \beta(A - U) = -S/k \tag{7.34}$$

Thus

$$S = -k < \ln P_r >= -k\sum_r P_r \ln P_r \tag{7.35}$$

This shows that the entropy is solely and completely determined by the probability P_r! When T is 0 K, P_r is equal to 1 for this state and 0 for the rest. Then the entropy is exactly 0. We also infer that the vanishing entropy and perfect statistical order go together. When the number of accessible states rises, the entropy of the system increases because more P_r appears.

For micro-canonical ensemble, $P_r = 1/\Omega$ Thus,

$$S = -k\sum_{r=1}^{\Omega}\frac{1}{\Omega}\ln(\frac{1}{\Omega}) = k\ln\Omega \tag{7.36}$$